Digital Curation Architecture for the Lunar Base Ecosystem

AE 8900 MAV Special Problems

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Lunar Extravehicular Activity: Enhanced Decision Making in a Virtual Environment

Executive Summary

This project, conducted under the Aerospace Systems Design Laboratory (ASDL) at the Georgia Institute of Technology and advised by Prof. Dimitri Mavris, explores the development of a digital curation architecture designed to enhance knowledge reuse, traceability, and interoperability across ASDL's Lunar Base ecosystem. As the laboratory advances its model-based systems engineering (MBSE) workflows, the need for a unified framework to organize, tag, and retrieve engineering artifacts has become increasingly critical.

The proposed system introduces a metadata-driven, AI-assisted approach to digital engineering knowledge management. Using the ASSESS ESMS 1.3 metadata standard [1], the framework employs structured way to characterize model identity, lifecycle stage, and utility across files of varying formats—ranging from requirements and CAD data to simulation scripts. These structured descriptors form the foundation for semantic retrieval and contextual search. By integrating LlamaIndex for local embedding generation and LM Studio for natural-language interaction, the system allows engineers to query their repository in human-readable terms while maintaining data privacy and offline accessibility.

The architecture supports the broader ASDL digital thread by connecting disparate model sources through metadata-based relationships rather than rigid schema alignment. This improves not only discoverability and reuse of past work but also the continuity of decision-making across project lifecycles. The expected outcome is a foundational curation capability that can later scale to support cross-project interoperability, predictive modeling, and automated knowledge graph construction within NASA's lunar habitat and infrastructure design ecosystem.

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Metadata-Driven Knowledge Management, ESMS 1.3, Digital Thread, LlamaIndex, AI-Assisted Curation

With the rapid expansion of digital engineering workflows within NASA's Lunar Base initiatives, ASDL's Model-Based Systems Engineering (MBSE) ecosystem faces the challenge of efficiently managing and reusing complex, interlinked artifacts. The Digital Curation Architecture introduced in this project provides a structured foundation for knowledge management and model traceability across the Pioneer and DELOS frameworks. By combining the ASSESS ESMS 1.3 metadata schema with AI-assisted retrieval through LlamaIndex, this system enables semantic tagging, contextual search, and lifecycle-aware organization of engineering data. The resulting framework enhances decision-making within the lunar habitat design process by bridging structured metadata with natural language accessibility—advancing the lab's broader digital thread integration for sustainable lunar exploration.

I. Introduction

The advancement of Model-Based Systems Engineering (MBSE) and Digital Thread initiatives within the Aerospace Systems Design Laboratory (ASDL) has led to a growing need for organized, interoperable, and reusable engineering data. Across projects such as DELOS, Pioneer, and LUMIA, ASDL researchers generate large volumes of heterogeneous artifacts—ranging from SysML models and MATLAB simulations to CAD files, Excel trade studies, and technical reports. While these models form the foundation of ASDL's Lunar Base ecosystem, their distributed and inconsistent organization presents persistent challenges to traceability, reuse, and collaborative decision-making.

To address these challenges, this project proposes the development of a Digital Curation Architecture for the Lunar Base Ecosystem. The system introduces a metadata-driven and AI-assisted framework for managing digital engineering artifacts, enabling structured storage, semantic search, and lifecycle awareness. It aims to provide a unified environment that bridges human workflows and machine reasoning, supporting the broader goal of establishing a continuous, intelligent digital thread for lunar mission design and analysis.

The architecture leverages the ASSESS Engineering Systems Modeling Standard (ESMS) 1.3, which defines six categories—Identity, Scope, Representation, Utility, Assessment, and Lifecycle Management—to describe engineering models in a machine-interpretable way. Each artifact is annotated using this metadata structure, encoded in YAML, ensuring consistent representation across the repository.

In parallel, the framework integrates AI-based semantic retrieval using LlamaIndex and LM Studio to enable natural-language querying and local vector-based search. This capability allows engineers to locate relevant models or documents through conversational interaction while maintaining data privacy and offline operation—a crucial requirement for secure and research-focused environments.

By aligning metadata standardization, graph-based relationships, and local AI tools, this architecture forms a foundation for scalable knowledge management within ASDL's digital engineering ecosystem. The ultimate goal is to enhance traceability, improve model reuse across projects, and support informed decision-making.

II. Background Review

A. Literature Review

The ASSESS Engineering Systems Modeling Standard (ESMS) 1.3 provides a formalized metadata framework designed to enhance the organization, interoperability, and reuse of engineering data across model-based systems engineering (MBSE) environments. It defines six key categories—Identity, Scope, Representation, Utility, Assessment, and Lifecycle Management—which collectively enable artifacts to be described in a consistent and machine-interpretable manner [1].

Encoding these metadata elements in YAML allows for structured tracking of not only a file's contents but also its purpose, stage of maturity, and relationship to other artifacts. This approach offers a semantic layer that bridges diverse engineering data types, ensuring traceability between different modeling efforts, including those under ASDL's DELOS and Pioneer initiatives.

Complementary research in graph-based digital curation demonstrates how repositories can be transformed into interconnected knowledge graphs rather than static file hierarchies [2]. Each artifact functions as a node, linked by semantic relationships such as *depends on* or *derived from*. This structure supports reasoning and discovery, enabling researchers to traverse design lineage, dependencies, and version history.

Finally, the integration of AI-assisted metadata tools such as LlamaIndex extends the reach of this framework by enabling natural-language retrieval and semantic embedding of heterogeneous artifacts [3]. Through local indexing and retrieval-augmented generation (RAG), engineers can interact with their repositories conversationally—asking, for example, "Show all thermal models used in DELOS Phase II"—while preserving data privacy and offline accessibility.

B. Research Gaps

Despite advances in MBSE tools and metadata standards, several persistent gaps limit efficient knowledge reuse within engineering research environments:

- ***Fragmented Repositories*** - Engineering data at ASDL is often distributed across multiple file systems, project drives, and formats (e.g., SysML, Excel, MATLAB, CAD). This fragmentation inhibits continuity between projects such as DELOS and Pioneer, where similar models and analyses are often recreated from scratch.

- ***Lack of Semantic Linkage*** - Traditional folder structures lack embedded semantic relationships. Without explicit metadata connections, it is difficult to understand how one artifact informs or depends on another—leading to loss of traceability and design rationale.

- ***Manual Curation and Limited Automation*** - Metadata assignment is often a manual, inconsistent process. There is currently no scalable, AI-assisted mechanism for identifying metadata fields or suggesting context-aware values based on document content.

- ***Insufficient Integration Between Metadata and AI Systems*** - While standards like ESMS provide structure, they lack direct mechanisms for human–AI interaction. There remains a gap between *metadata definition* and *metadata utilization*, particularly in how AI models can dynamically query or reason over structured engineering repositories.

These gaps underscore the need for a flexible, AI-assisted digital curation system that combines metadata consistency, semantic reasoning, and natural-language interaction.

C. Research Objective

The goal of this research is to design and prototype a Digital Curation Architecture for the Lunar Base Ecosystem, integrating standardized metadata, graph-based linkage, and AI-driven retrieval. The specific objectives are:

- ***Implement Metadata Standardization*** - Apply the ESMS 1.3 framework to create a uniform YAML-based metadata structure for all Lunar Base artifacts, ensuring consistent characterization of file identity, lifecycle stage, and intended use.

- ***Develop a Semantic Curation Layer*** - Enable cross-linking of engineering artifacts through graph-based relationships that capture dependencies, lineage, and subsystem interactions across DELOS, Pioneer, and LUMIA models.

- ***Integrate AI-Assisted Indexing and Retrieval*** - Utilize LlamaIndex and LM Studio for local embedding generation, allowing engineers to perform context-aware, natural-language queries without requiring external API calls or cloud-based services.

- ***Enhance Knowledge Reuse and Decision Support*** - Facilitate faster discovery of relevant models, trade studies, and simulations, strengthening traceability and supporting informed decision-making across project lifecycles.

Collectively, these objectives aim to establish the foundation for a sustainable digital thread within ASDL—one that supports scalable, intelligent, and privacy-preserving knowledge management for lunar mission architecture development.

III. Operational Scenario

The proposed digital curation architecture functions as a modular, self-contained system designed to unify and contextualize the diverse set of engineering artifacts that make up ASDL's Lunar Base ecosystem. At its core, the architecture integrates metadata standardization, semantic indexing, and user interaction into a cohesive workflow. The metadata layer ensures that all files—ranging from trade studies and thermal models to control algorithms—adhere to the ASSESS ESMS 1.3 standard, thereby maintaining semantic consistency across disciplines. Each file is annotated with structured YAML metadata reflecting its identity, scope, representation, utility, assessment, and lifecycle state. These descriptors allow the system to understand the role and relevance of each artifact within the broader design ecosystem. Building upon this foundation, the semantic indexing layer, powered by LlamaIndex, embeds each artifact's textual and structural features into a vector space, enabling nuanced similarity searches and contextual retrieval. The final layer leverages LM Studio, which provides a local, conversational interface through which

users can query, navigate, and reason about the curated repository. Together, these layers form a scalable and privacy-preserving system that bridges human intuition with machine reasoning.

The operational workflow begins with the ingestion of artifacts from existing project directories within ASDL's research infrastructure. These may include datasets from DELOS trade studies, CAD files from the Pioneer habitat design effort, or simulation codes from thermal subsystem analyses. As each file is processed, the system automatically identifies its type, reads relevant textual or numerical content, and populates the corresponding YAML metadata template. For instance, a thermal analysis Python script might be labeled under *Representation: Thermal* and *Lifecycle: Active*, while a DELOS spreadsheet summarizing mass results may be assigned *Scope: Outputs* and *Assessment: Valid*. When the system encounters incomplete or ambiguous information, it invokes a locally running large language model through LM Studio to propose metadata suggestions based on contextual understanding. This human-in-the-loop tagging process blends automation and expert review, ensuring both efficiency and accuracy. Over time, the metadata repository evolves into a living catalog of engineering knowledge that captures not only the files themselves but also their relationships, assumptions, and maturity.

Once the repository is indexed, users interact with it through a natural-language search interface. Rather than navigating nested folders or memorizing filenames, an engineer can type or speak queries such as *"Show all ISRU power models developed after April 2024"* or *"Find trade studies related to thermal mass optimization."* LlamaIndex converts these queries into vector embeddings and retrieves semantically relevant artifacts from the local index, ranking them by contextual similarity. The interface presents each result alongside its metadata summary, including key ESMS tags, file type, and last modification date, allowing immediate interpretability and traceability. This capability enables engineers to query by intent—whether conceptual, functional, or temporal—rather than relying on rigid keyword searches. As a result, researchers can explore the repository as a connected graph of knowledge rather than a static archive of files.

A practical example of this workflow can be seen in the cross-project reuse of thermal system models. Suppose a thermal subsystem engineer working on the Pioneer habitat seeks to build upon previous work from the DELOS mission. By querying *"Retrieve radiator sizing codes and DOE studies related to lunar surface operations,"* the system identifies several key artifacts:

radiators_24_Grand_Challenge.py (a radiator sizing model), *DoE_DELOS_Final_240404.xlsx* (a trade study of energy configurations), and *Important Equations.docx* (a document containing analytical derivations). Together, these results reconstruct the lineage of an earlier modeling effort—from equations to code to data—enabling the engineer to trace assumptions, validate parameters, and adapt proven methodologies to the new design context. This interaction exemplifies how the digital curation architecture enhances traceability, preserves institutional memory, and accelerates knowledge transfer across ASDL's research portfolio.

Ultimately, this operational scenario demonstrates how metadata standardization, local semantic search, and AI-assisted curation converge to transform the traditional engineering repository into a living knowledge system. The architecture not only reduces redundancy and retrieval time but also strengthens continuity across project generations, forming a cornerstone for the laboratory's broader vision of an intelligent, model-centric lunar base design ecosystem.

IV. Methodology

The The curation prototype adopts a retrieval-augmented generation (RAG) architecture in which documents and their ESMS-aligned YAML descriptors are ingested, chunked into semantic "nodes," embedded, and persisted in a local vector store for low-latency retrieval and synthesis. Concretely, the pipeline is built around LlamaIndex's VectorStoreIndex and vector-store integrations. After files are discovered in the Lunar Base repository (e.g., DELOS trade studies, Pioneer subsystem analyses, thermal codes), a loader parses their contents into LlamaIndex Document objects. Each document is then transformed into smaller, overlap-aware chunks via the node parser, which improves retrieval granularity and mitigates context dilution during synthesis. These nodes are embedded using a local embedding model and written into a persistent vector store (Chroma in the baseline configuration), providing a durable, on-disk index that can be reloaded without full re-ingestion. The index exposes a retriever abstraction that resolves user queries to top-k nodes using approximate nearest-neighbor similarity, and a query engine that composes retrieved context into grounded responses. This arrangement mirrors the recommended LlamaIndex flow—VectorStoreIndex atop a pluggable store and a retriever/query-engine interface—while keeping all data and computation local for privacy and reproducibility.

ESMS metadata is integrated at ingestion to enrich retrieval with structure and intent. YAML fields—such as Representation (e.g., “thermal”), Scope (“inputs,” “outputs”), Lifecycle Management (“maintained,” “active”), and Utility (“trade study,” “validation”)—are attached as per-node metadata and carried into the vector store. During retrieval, these tags are used as filters and re-ranking signals so that queries like “radiator sizing models with maintained lifecycle after April 2024” narrow the candidate set before similarity search. Because LlamaIndex treats vector stores as first-class, the same codepath supports other local or self-hosted stores (FAISS, LanceDB, Qdrant, etc.) with only configuration changes; we selected Chroma for persistence and simple metadata filtering, but the design remains store-agnostic for future scaling.

To operationalize the workflow, a thin Python layer orchestrates four steps: (1) ingestion and parsing of repository files and their YAML; (2) chunking and embedding into a VectorStoreIndex; (3) persistence of the store to disk for incremental updates; and (4) retrieval and synthesis via a query engine that accepts natural-language prompts from a local interface (LM Studio). At query time, the retriever applies optional ESMS filters (e.g., representation == "thermal" and lifecycle == "maintained"), runs vector similarity to fetch the most relevant nodes, and hands them to the response synthesizer. This produces answers grounded in the underlying artifacts (spreadsheets, scripts, reports) rather than the model’s prior, aligning with best practices for explainable, auditable RAG. The overall method follows LlamaIndex’s documented patterns for vector indexing, store persistence, metadata filters, and retrieval composition, while remaining entirely offline.

V. Technical Approach

The technical approach implements the RAG-based methodology as a fully local, modular stack that couples ESMS-aligned metadata with embedding search and conversational access. The system runs on Python 3.11 and is organized around three cooperating layers: a curation layer that produces YAML metadata for every artifact, an indexing layer that transforms both documents and metadata into a persistent vector store, and an interaction layer that exposes retrieval and synthesis through a local LLM. All processing remains on the workstation to preserve data sovereignty and to enable reproducible, offline operation.

At the curation layer, every file in the Lunar Base repository is paired with a YAML record conforming to ESMS 1.3, capturing identity, scope, representation, utility, assessment, and lifecycle details. These YAML descriptors are stored beside the source data and act as the system's "schema of meaning." During ingestion, a lightweight parser reads both the raw artifact and its YAML, normalizes fields, and attaches ESMS attributes as node-level metadata.

The indexing layer follows the LlamaIndex vector-stores pattern. Each artifact and its YAML are converted into Document objects, chunked into overlap-aware nodes to preserve local context, and embedded with a local sentence-embedding model. Nodes and their ESMS metadata are written to a persistent vector store (Chroma in the baseline configuration, with FAISS or Qdrant as drop-in alternatives), allowing the index to be reopened quickly between sessions without re-processing the entire repo. Because ESMS keys are stored as metadata on every node, retrieval can combine dense similarity with structural filtering; a query such as "radiator thermal sizing models with maintained lifecycle" first restricts candidates by representation=thermal and lifecycle=maintained, then ranks results by vector similarity, which yields fast, precise matches across formats.

The interaction layer exposes two local interfaces tuned to everyday use. For conversational retrieval, LM Studio runs an open-weights model that receives a user's natural-language question together with context retrieved by LlamaIndex's retriever; the query engine assembles a grounded response using only the returned nodes, and includes short citations (file titles and relative paths) for traceability. For programmatic workflows, a thin FastAPI service wraps common operations—ingest, reindex, search, and summarize—so engineers can script repeatable tasks or integrate retrieval into analysis notebooks. All requests, retrieved node IDs, and response summaries are logged to a lightweight SQLite file to support auditability, ablation testing, and future error analysis.

Several practical choices make the system robust in ASDL's environment. Embedding and tokenization run locally to avoid network dependencies; batch sizes and chunk overlap are tuned to keep GPU-less laptops responsive; and the vector store is persisted under a project-scoped directory, with simple versioning that snapshots index state before large updates. YAML validation occurs on write and again at index time so malformed records never enter the store.

Finally, the stack is intentionally swappable: the embeddings model, vector store, and chat model are each abstracted behind configuration flags, allowing upgrades (for accuracy or performance) without changes to the higher-level workflow.

Together these components realize a compact but capable technical solution: ESMS gives the repository a consistent semantic spine; LlamaIndex fuses that structure with content-level embeddings in a durable index; and LM Studio provides private, conversational access to grounded results. The outcome is a local, reproducible system that turns a conventional file tree into an immediately searchable, explainable, and extensible knowledge resource for the Lunar Base program.

VI. Results and Discussion

VII. Conclusion and Future Work

VIII. References

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